Software Project Management
Assignment-5
TYPE OF QUESTION: MCQ/MSQ

Number of questions: **10** **Total mark: 10 X 1 = 10**

QUESTION 1:

Which of the followings are NOT examples of expert judgment-based technique?

a. Basic expert judgement
b. Weighted average estimating
c. Delphi cost estimation
d. COCOMO
e. Consensus estimating
f. Halstead's software science

Correct Answer: d. COCOMO
f. Halstead's software science

Detailed Solution:

Expert judgment techniques rely on human expertise, experience, and consensus—examples include basic expert judgement, weighted average estimating, Delphi, and consensus estimating. COCOMO is an algorithmic model based on mathematical formulas, and Halstead's software science is an analytical metric based on operators and operands; neither uses expert judgment as their primary mechanism.

QUESTION 2:

Which of the followings are examples of organic project?

a. Accounting system
b. Compilers
c. Operating systems
d. Real-time systems
e. Student Information System
f. Linkers

Correct Answer: a. Accounting system
e. Student Information System

Detailed Solution:

In the COCOMO model, organic projects are small, simple, and well-understood, with experienced teams and flexible requirements—typical examples include accounting systems and student information systems. Compilers and linkers are semi-detached, while operating systems and real-time systems are embedded due to their complexity and tight constraints.

QUESTION 3:

In intermediate COCOMO, if there are stringent reliability requirements on the product, then the initial cost and effort estimates obtained by the basic COCOMO, are

a. scaled downwards
b. scaled upwards
c. remained unchanged
d. potentially decreased
e. halved

Correct Answer: b. scaled upwards

Detailed Solution:

In intermediate COCOMO, the cost driver RELY (required software reliability) has a value greater than 1.0 for stringent reliability requirements (e.g., High = 1.15, Very High = 1.30). This multiplies the basic COCOMO estimate by an Effort Adjustment Factor (EAF) > 1.0, thus scaling the effort and cost upwards—not downward, unchanged, decreased, or halved.

QUESTION 4:

A distributed decision support System (DSS) for an organization having offices at several places across the country has the following modules: a database module, a graphical user interface (GUI) module and a communication module. Which of the following COCOMO models will be best suitable for estimating the development cost?

a. Basic COCOMO
b. Intermediate COCOMO
c. COCOMO 81
d. Detailed (Complete) COCOMO
e. All of these

Correct Answer: d. Detailed (Complete) COCOMO

Detailed Solution:

A distributed DSS with database, GUI, and communication modules involves different types of components, each with its own complexity, cost drivers, and development environment. The Detailed (Complete) COCOMO model is best suited because it allows estimating effort module by module (with phase-level and module-level cost drivers) and then summing them, providing greater accuracy than Basic or Intermediate COCOMO, which apply uniform cost drivers to the entire project.

QUESTION 5:

Which one of the following COCOMO II sub-models would be used when requirements are available but design has not yet started?

a. Application composition
b. Early design
c. Reuse
d. Post-architecture
e. Application configuration

Correct Answer: b. Early design

Detailed Solution:

The Early design sub-model of COCOMO II is specifically used when requirements are known but the detailed software design and architecture have not yet been completed. It uses fewer cost drivers (7 instead of 17) and provides a coarse estimate based on function points or KSLOC. Application composition is for very early prototyping, reuse is for estimating effort to integrate reused components, and post-architecture applies once the architecture is fully defined.

QUESTION 6:

Post-architecture model of COCOMO II is primarily used for

a. systems developed using dynamic languages, DB programming etc.
b. initial effort estimation based on system requirements and design options
c. computing the effort to integrate the reusable components or automatically generated code
d. estimating the development effort based on system design specification
e. Evaluating the impact of project cost drivers and scale factors on the final effort estimate

Correct Answer: d. estimating the development effort based on system design specification
e. Evaluating the impact of project cost drivers and scale factors on the final effort estimate

Detailed Solution:

The Post-Architecture model of COCOMO II is primarily used for estimating development effort based on the system design specification (d), as it leverages detailed size inputs and 17 cost drivers once the architecture is stable. Additionally, it serves to evaluate the impact of project cost drivers and scale factors (e) by allowing managers to adjust parameters like team capability or reliability requirements and observe their effect on the final effort estimate, making both options correct and complementary uses of the model.

QUESTION 7:

Which of the following estimation techniques are based on making an educated guess of the project parameters?

a. Expert judgement
b. Basic COCOMO
c. Halstead's software science
d. Delphi cost estimation
e. Intermediate COCOMO
f. Complete COCOMO

Correct Answer: a. Expert judgement
d. Delphi cost estimation

Detailed Solution:

Expert judgement and Delphi cost estimation rely on human expertise, experience, and consensus to make informed "educated guesses" about project parameters. In contrast, Basic COCOMO, Intermediate COCOMO, Complete COCOMO, and Halstead's software science are algorithmic or analytical models that use fixed mathematical formulas or metrics, not subjective human estimation.

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QUESTION 8:

In Halstead's software science,

a. the programming effort varies inversely with the volume.
b. the programming effort varies linearly with the volume.
c. the programming effort varies as the square of the volume.
d. the programming effort varies as the cube of the volume.
e. the programming effort does not vary with the volume.

Correct Answer: c. the programming effort varies as the square of the volume.

Detailed Solution:

In Halstead's software science, programming effort (E) is proportional to the square of the program volume (E∝V2E∝V^2), where volume (VV) measures the information content of the program based on operators and operands. Effort does not vary inversely, linearly, as the cube, or remain unaffected by volume—the quadratic relationship is a core result of Halstead's metrics.

QUESTION 9:

Which one of the following is NOT a reason for late project becoming later due to addition of manpower?

a. Increased communication among team members
b. Learning by the added members
c. Increased project scope
d. Limited scope for parallelizing various project activities
e. Additional effort required from experienced team members to train and mentor new members

Correct Answer: c. Increased project scope

Detailed Solution:

Brooks' Law states that adding manpower to a late project makes it later due to factors like increased communication overhead, learning curves for new members, limited parallelizability of tasks, and training efforts required from experienced staff. Increased project scope, however, is not a consequence of adding manpower—it is a separate decision (scope creep) and not a reason why additional people delay the project.

QUESTION 10:

Suppose the development time of a small software product has been estimated to be one and half year. Then, in order to develop the product in six months, what will be total increase in development effort? (Use Jensen's model).

a. 3 times
b. 6 times
c. 9 times
d. 12 times
e. 18 times
f. 27 times
g. 81 times

Correct Answer: c. 9 times

Detailed Solution:

Jensen's model (based on Putnam's software equation) relates effort (E) and time (T) as $E \propto 1/T^2$ for a fixed product size. Reducing the schedule from 1.5 years (18 months) to 0.5 years (6 months) gives a time ratio of 18/6=318/6=3. Thus, the effort increases by 3^2=9 times.

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